Technical Documentation: Design, Construction, and Analysis of a Batch-Type Alkaline Water Electrolysis Reactor

Project Team: Infinity Explorers

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Abstract

This document provides a comprehensive overview of a rudimentary, unipolar Alkaline Water Electrolysis (AWE) batch reactor. Part I details the practical assembly and material requirements for constructing a prototype. Part II provides a rigorous engineering and electrochemical analysis of the system, detailing the thermodynamics, reaction kinetics, material degradation modes, efficiency calculations, and safety architecture involved in the stoichiometric production of oxyhydrogen gas.

Part I: System Assembly and Construction Manual

1. Introduction

Water (H_2O) is composed of two parts hydrogen and one part oxygen. Hydrogen is a highly energy-dense, flammable substance, while oxygen is a strong oxidizer that supports combustion. To harness this stored chemical energy, the molecular bonds must be cleaved using electrical work—a process known as electrolysis. This system splits water into hydrogen and oxygen gases, producing a highly combustible stoichiometric mixture (HHO) applicable for small-scale experimental power demonstrations.

2. Bill of Materials (BOM)

- **Primary Reactor Vessel:** Transparent plastic container with airtight lid (~900ml)
- **Flashback Arrestor Vessel:** Transparent plastic container (~100ml)
- **Electrodes:** Hex Bolt 2.5 / 6 Soot (2 units)

- **Hardware:** Hex Nut 6mm (24 units), Washer M6 (18 units)
- **Pneumatic Routing:** Plastic tubing (1 meter), Plastic pipe (15 cm), Hose barb
- **Electrical:** 9V DC battery, 9V connector, DC Switch, Insulated copper wires
- **Sealing:** Rubber/foam sheet, Epoxy glue, Super glue, Cellophane tape, Electrical tape
- **Electrolyte:** Potassium Hydroxide (KOH) or Sodium Bicarbonate ($NaHCO_3$)
- **Solvent:** Distilled Water

3. Required Tools

Cutter, Pliers, Wrench, Sandpaper.

4. Preparations

1. **Sanitization:** Clean both the primary and secondary plastic containers with surfactant (dishwashing liquid) and water. Ensure absolute dryness.
2. **Surface Preparation:** Inspect nuts, bolts, and washers for oxidation (rust). Remove any passivated oxide layers using rust remover or mechanical abrasion (sandpaper). The electrode surfaces must be exposed, bare metal to reduce contact resistance.

5. Assembly Procedure

Step 1: Electrode Fabrication Thread a washer and a nut onto a hex bolt, fastening tightly with a wrench. Repeat this sequence 9 times per bolt to maximize the localized surface area, which helps lower the current density and subsequently the activation overpotential. Assemble two identical units to serve as the anode and cathode.

Step 2: Pneumatic Sealing (Arrestor) Trace the lid of the secondary container onto the rubber/foam sheet. Cut a ring slightly smaller than the trace and insert it under the cap to serve as an O-ring, ensuring a hermetic seal.

Step 3: Vessel Machining Puncture three precision holes into the primary reactor lid. One hole accommodates the pneumatic hose barb; the other two secure the electrode bolts. The fit must be strictly interference-based (tight) to prevent gas leaks and pressure loss. Puncture two holes in the secondary container lid for the inlet/outlet pipes.

Step 4: Reactor Integration Apply epoxy resin to the primary lid holes. Seat the hose barb. Insert the electrode assemblies through the lid, securing them with nuts on both the internal and external faces, compressing the lid material to form a seal. Leave adequate thread length exposed externally for electrical terminals.

Step 5 & 6: Arrestor Plumbing Cut the rigid plastic pipe into two unequal lengths. The inlet pipe must extend from above the lid to near the bottom of the container (submerged). The outlet pipe should barely penetrate the lid (unsubmerged). Secure both with epoxy resin.

Step 7: Electrical Circuit Wire the 9V battery connector in series with the DC switch. Connect the switch output and the negative battery terminal to the exposed electrode threads. Insulate all exposed copper with electrical tape. Affix the power supply to the reactor chassis using super glue and cellophane tape.

Step 8: Electrolyte Charging Fill the primary reactor with distilled water and dissolve 1 teaspoon of Potassium Hydroxide (KOH) or Sodium Bicarbonate ($NaHCO_3$) to create the ionic electrolyte. Seal tightly. Fill the secondary container (arrestor) with distilled water, ensuring the inlet pipe is submerged, but keeping the water level below the outlet pipe.

Step 9: Final Connection With the switch in the OPEN (OFF) state, secure the terminal wires to the electrodes using the uppermost nuts. Do not solder, as the system requires periodic disassembly for fluid replenishment. Connect the pneumatic tubing from the primary reactor output to the submerged inlet of the arrestor. Output gas will be harvested from the arrestor's unsubmerged outlet.

6. Safety and Precautions

- **Explosion Hazard:** The effluent gas is a stoichiometric mixture of H_2 and O_2 (Brown's Gas), which is extremely volatile and possesses a very low ignition energy (0.02 mJ). Keep away from sparks, open flames, and electrostatic discharges.
- **Storage Prohibition:** Never compress or store the generated oxyhydrogen gas. Generate strictly on-demand.
- **Isolation:** Disconnect the power source physically when not in active use.
- **PPE:** Chemical splash goggles and nitrile gloves are mandatory, particularly when handling KOH (a caustic alkali).
- **Ventilation:** Operate exclusively in well-ventilated areas or fume hoods.

Part II: Advanced Electrochemical & Engineering Analysis

While the mechanical assembly yields a functional prototype, evaluating the system through the lens of electrochemical engineering reveals significant thermodynamic inefficiencies and material limitations. The following sections detail the mathematical and physical parameters governing the reactor's behavior.

1. Electrochemical Thermodynamics

The core principle is the forced non-spontaneous dissociation of water. The overall endothermic reaction is:

$$H_2O(l) \rightarrow H_2(g) + \frac{1}{2}O_2(g)$$

Under standard conditions ($T = 298\ \mathrm{K}$, $P = 1\ \mathrm{atm}$), the thermodynamic energy requirements are defined by the change in Gibbs free energy ($\Delta G^\circ = 237.1\ \mathrm{kJ/mol}$) and Enthalpy ($\Delta H^\circ = 285.8\ \mathrm{kJ/mol}$).

The theoretical minimum voltage (Reversible Cell Potential, E°_{rev}) to drive this reaction without heat exchange is derived via the Nernst equation framework:

$$E^\circ_{rev} = \frac{\Delta G^\circ}{zF} \approx 1.23 \text{ V}$$

Where:

- $z = 2$ (moles of electrons transferred per mole of H_2)
- $F = 96485 \text{ C/mol}$ (Faraday's constant)

However, operating at 1.23 V requires the cell to absorb ambient thermal energy ($T\Delta S$). To operate *thermoneutrally* (sustaining thermal equilibrium without external heating or cooling), the applied voltage must equal the thermoneutral voltage (E_{th}):

$$E_{th} = \frac{\Delta H^\circ}{zF} \approx 1.48 \text{ V}$$

2. Half-Cell Kinetics and Overpotentials

The prototype utilizes Potassium Hydroxide (KOH), creating an alkaline electrolyte system which provides essential OH^- ions for conductivity. High pH environments ($pH > 13$) are favorable for avoiding the use of expensive noble metals.

Alkaline Half-Cell Reactions:

- **Cathode (Reduction):** $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq) \quad (E^\circ = -0.828 \text{ V})$
- **Anode (Oxidation):** $2OH^-(aq) \rightarrow \frac{1}{2}O_2(g) + H_2O(l) + 2e^- \quad (E^\circ = 0.401 \text{ V})$

In practice, applying 1.48 V yields negligible current due to kinetic barriers known as overpotentials (η). The actual operating cell voltage (V_{cell}) is expressed as:

$$V_{cell} = E^\circ_{rev} + \eta_{act,a} + |\eta_{act,c}| + \eta_{ohm} + \eta_{conc}$$

The steel hex bolts utilized possess poor electrocatalytic activity. This results in a massive **activation overpotential** (η_{act}), governed by the Tafel equation:

$$\eta_{act} = \frac{RT}{\alpha zF} \ln\left(\frac{i}{i_0}\right)$$

Where i_0 is the exchange current density. For bare steel, i_0 is exceptionally low, requiring a severe voltage penalty to achieve macroscopic gas evolution. Furthermore, the distance between the two bolts (the inter-electrode gap) significantly increases the ohmic overpotential (η_{ohm}) due to the resistance of the aqueous electrolyte.

3. System Architecture & Material Failure Modes

A. Anodic Dissolution (Corrosion)

The most critical failure vector in this design is the use of unprotected low-carbon steel for the anode. In an alkaline environment under extreme oxidizing potentials, the oxygen evolution reaction (OER) is outcompeted by the anodic dissolution of iron:

$$Fe(s) \rightarrow Fe^{2+}(aq) + 2e^{-}$$

$$Fe^{2+}(aq) + 2OH^{-}(aq) \rightarrow Fe(OH)_2(s)$$

This parasitic oxidation will rapidly consume the anode, precipitating iron hydroxides/oxides (rust) into the electrolyte. This sludge increases ohmic resistance and eventually halts the reaction. Industrial systems circumvent this by utilizing Nickel or Nickel-plated steel, which forms a stable, passivating $NiOOH$ layer in alkaline conditions, highly active for OER while resisting corrosion.

B. Power Supply Inefficiency

The utilization of a 9V DC source on a single unipolar cell introduces severe ohmic losses. Because the thermoneutral threshold is $\sim 1.48\ \mathrm{V}$, applying $9\ \mathrm{V}$ across a single electrolyte gap means approximately $7.5\ \mathrm{V}$ is dissipated entirely as Joule heating (I^2R losses), which warms up the water rather than splitting it.

- **Voltage Efficiency:** $\varepsilon_V = \frac{1.48\ \mathrm{V}}{9.0\ \mathrm{V}} \approx 16.4\%$

Commercial electrolyzers employ bipolar stack architectures—arranging multiple thin cells in series—to divide the source voltage efficiently across numerous gaps, thereby maximizing energy utilization.

C. Fluidic Safety: The Hydraulic Flame Arrestor

The secondary container ("bubbler") acts as a critical safety device known as a **Hydraulic Flame Arrestor** or wet scrubber. Because the prototype lacks a semi-permeable membrane to separate the half-cells, the evolved gases immediately mix, creating oxyhydrogen (HHO). Oxyhydrogen has a wide flammability limit and a detonation velocity exceeding $2800\ \mathrm{m/s}$. The bubbler ensures that if the output gas is ignited, the resultant flame front propagating backward down the tubing is quenched by the liquid water column, cooling the gas below its autoignition temperature and preventing the detonation of the primary reactor volume.

4. Mass Transfer, Faradaic Efficiency, and Production Yield

The theoretical volumetric yield is strictly governed by Faraday's First Law of Electrolysis. The mass (m) of hydrogen produced correlates directly to the total integral of electrical charge (Q):

$$m = \frac{Q \cdot M}{z \cdot F} = \frac{\int I dt \cdot M}{z \cdot F}$$

Assuming a constant current, the theoretical volume of Hydrogen gas produced at Standard Temperature and Pressure (STP: $273.15\ \mathrm{K}$, $1\ \mathrm{atm}$) can be approximated using the ideal gas law ($1\ \mathrm{mole} = 22.4\ \mathrm{L}$):

$$V_{H2,theoretical} = \left(\frac{I \cdot t}{2 \cdot 96485} \right) \cdot 22.4 \text{ Liters}$$

Faradaic Efficiency (η_F): In this DIY build, the actual yield will be lower than the theoretical yield. Faradaic efficiency is the ratio of actual gas produced to the theoretical limit:

$$\eta_F = \frac{V_{H2,actual}}{V_{H2,theoretical}} \times 100\%$$

Because a portion of the electrical current is wasted on side reactions (specifically the corrosion/dissolving of the iron anode, as well as shunt currents in the fluid), the Faradaic efficiency of this system will likely fall below 80%.

5. Industrial Scalability & Membrane Integration

To graduate from a dangerous, mixed-gas DIY setup to a professional, scalable energy storage system, future iterations must incorporate a separator. Modern industrial AWE systems utilize porous diaphragms (like Agfa's Zirfon) or Proton Exchange Membranes (PEM, like DuPont's Nafion). These ion-permeable membranes physically divide the reactor into two distinct chambers, isolating the Hydrogen and Oxygen streams directly at the point of evolution, thereby ensuring pure gas outputs (up to 99.999% purity) and completely eliminating the explosion risk associated with mixed oxyhydrogen systems.